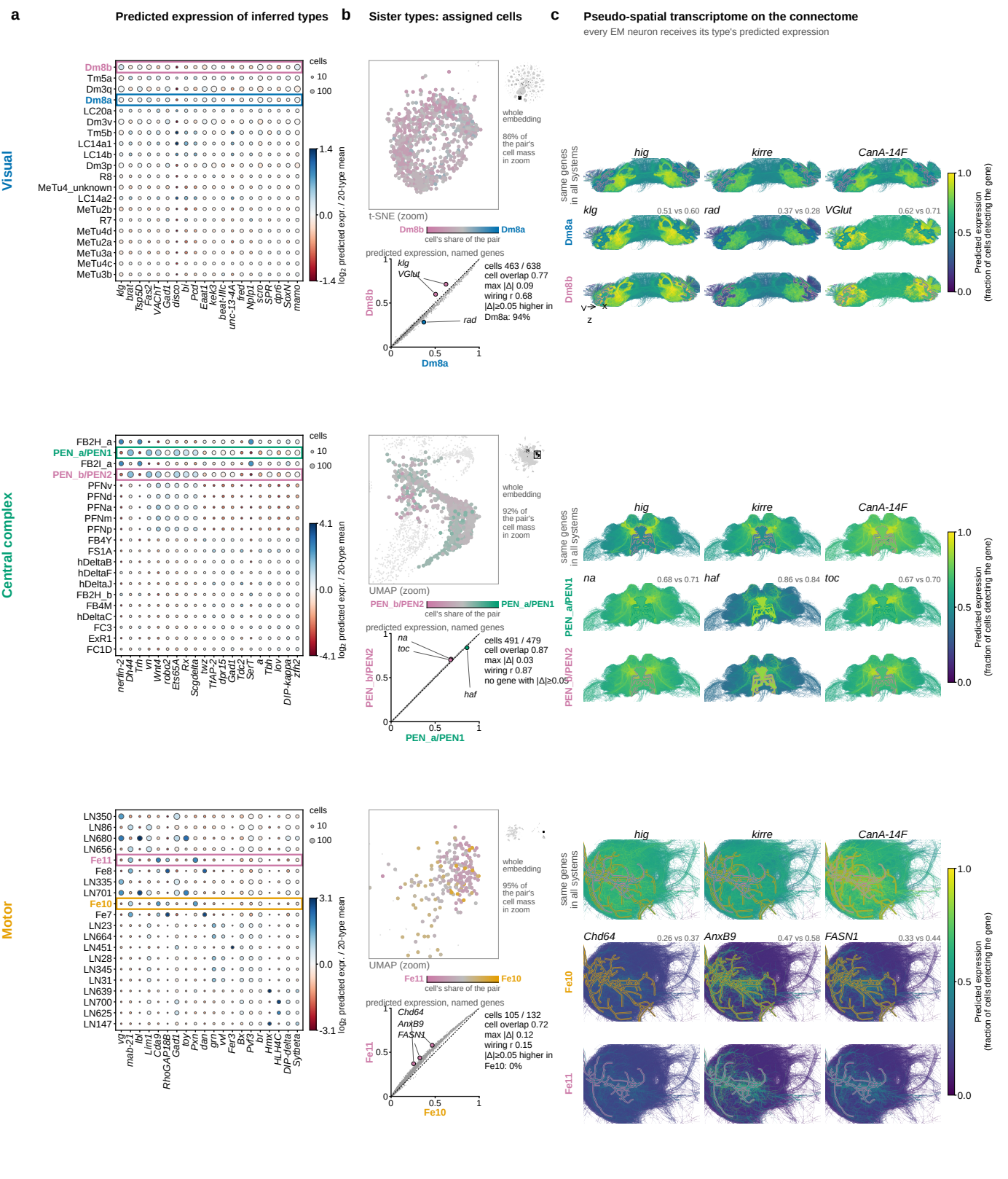

a
Predicted expression of inferred types
b
Sister types: assigned cells
c
Pseudo-spatial transcriptome on the connectome
every EM neuron receives its type's predicted expression
Visual
Central complex
Motor
cells
10
100
log2 predicted expr. / 20-type mean
Dm8b
Dm8a
t-SNE (zoom)
whole embedding
86% of the pair's cell mass in zoom
cell's share of the pair
predicted expression, named genes
cells 463 / 638
cell overlap 0.77
max |Δ| 0.09
wiring r 0.68
|Δ|≥0.05 higher in Dm8a: 94%
same genes in all systems
hig
kirre
CanA-14F
klg
0.51 vs 0.60
rad
0.37 vs 0.28
VGlut
0.62 vs 0.71
Predicted expression (fraction of cells detecting the gene)
PEN_a/PEN1
PEN_b/PEN2
UMAP (zoom)
95% of the pair's cell mass in zoom
na
0.68 vs 0.71
haf
0.86 vs 0.84
toc
0.67 vs 0.70
cells 491 / 479
cell overlap 0.87
max |Δ| 0.03
wiring r 0.87
no gene with |Δ|≥0.05
Fe11
Fe10
Chd64
0.26 vs 0.37
AnxB9
0.47 vs 0.58
FASN1
0.33 vs 0.44
cells 105 / 132
cell overlap 0.72
max |Δ| 0.12
wiring r 0.15
|Δ|≥0.05 higher in Fe10: 0%